



Telco Customer Churn Analysis Dashboard

1.95K

Total Customers

813

Churned Customers

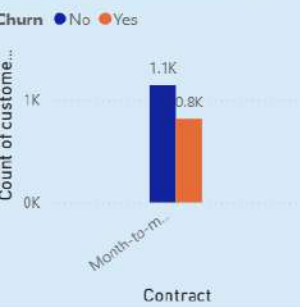
41.69%

Churn Rate %

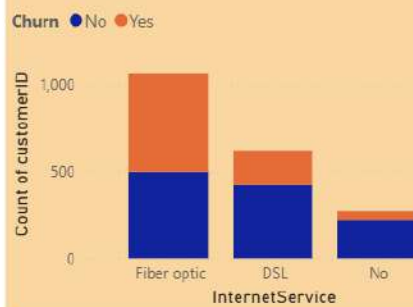
66.15

Avg Monthly Charges

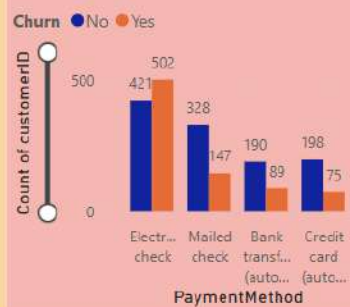
Customer churn by Contract



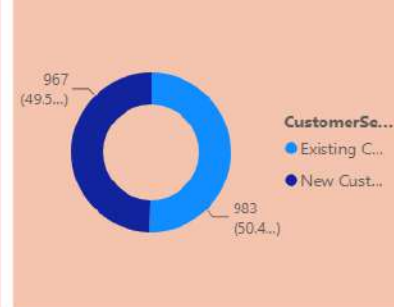
Churn By Internet Services



Churn By Payment Method



CustomerSegment



Contract

- Month-to-month
- One year
- Two year

InternetService

- DSL
- Fiber optic
- No

gender

- Female
- Male



Telco Customer Churn Analysis Dashboard

3.488K

Total Customers

939

Churned Customers

26.92%

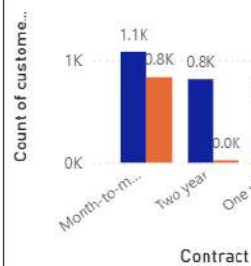
Churn Rate %

65.20

Avg Monthly Charges

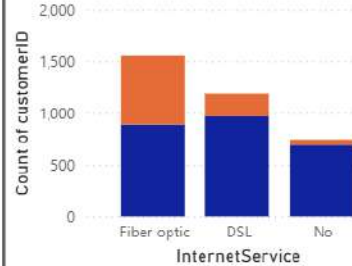
customer churn by Contract

Churn ● No ● Yes



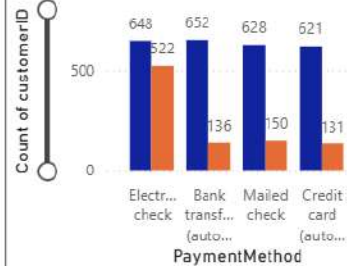
Churn By Internet Services

Churn ● No ● Yes

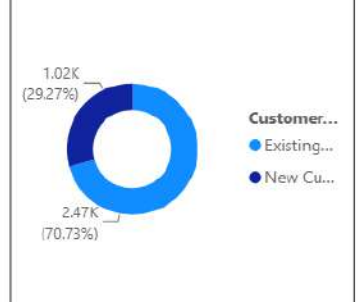


Churn By Payment Method

Churn ● No ● Yes



CustomerSegment



Contract

- ☐ Month-to-month
- ☐ One year
- ☐ Two year

InternetService

- ☐ DSL
- ☐ Fiber optic
- ☐ No

gender

- ☒ Female
- ☐ Male



Telco Customer Churn Analysis Dashboard

7.046K

Total Customers

1.869K

Churned Customers

26.53%

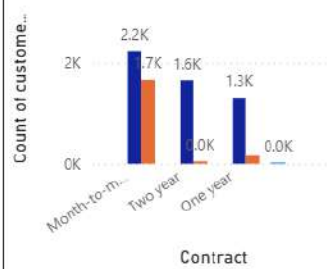
Churn Rate %

64.76

Avg Monthly Charges

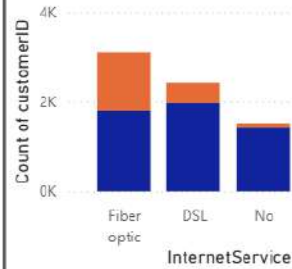
customer churn by Contract

Churn ● No ● Yes



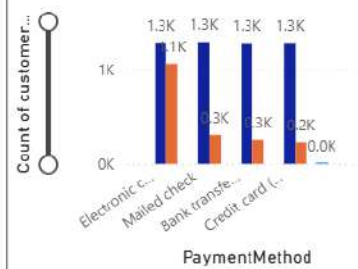
Churn By Internet Services

Churn ● No ● Yes

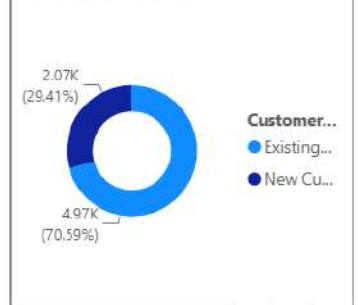


Churn By Payment Method

Churn ● No ● Yes



CustomerSegment



Contract

- ☐ Month-to-month
- ☐ One year
- ☐ Two year

InternetService

- ☐ DSL
- ☐ Fiber optic
- ☐ No

gender

- ☐ Female
- ☐ Male

architmehta0 / Telco-Customer-Churn-Analysis

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Quick setup — if you've done this kind of thing before

Set up in Desktop

 or

HTTPS

SSH

https://github.com/architmehta0/Telco-Customer-Churn-Analysis.git

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

```
echo "# Telco-Customer-Churn-Analysis" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/architmehta0/Telco-Customer-Churn-Analysis.git
git push -u origin main
```

Telco-Customer-Churn-Analysis / README.md in main

Cancel changes Commit changes...

Edit Preview

Spaces 2 No wrap

```
1 # Telco Customer Churn Analysis
2
3 ## Project Overview
4
5 This project analyzes customer churn data from a telecommunications company to identify factors influencing customer attrition. The analysis was performed using Excel, SQL, Python, and Power BI to generate actionable insights.
6
7 ## Problem Statement
8
9 Customer churn is a major challenge for telecom companies because losing customers directly impacts revenue and growth. The objective of this project is to analyze customer behavior, identify churn patterns, and provide recommendations to reduce churn.
10
11 ## Tools Used
12
13 * Microsoft Excel
14 * SQL
15 * Python
16 * Power BI
17 * VS Code
18
19 ## Project Workflow
20
21 1. Data Cleaning using Excel
22 2. Data Exploration using SQL
```

Use **Control + Shift + M** to toggle the **Tab** key moving focus. Alternatively, use **esc** then **Tab** to move to the next interactive element on the page.

Attach files by dragging & dropping, selecting or pasting them.

```
# Telco Customer Churn Analysis

## Project Objective

#Analyze customer churn using Excel, SQL, and Python.
```

Step 1: Load Dataset

```
import pandas as pd
import numpy as np

df = pd.read_csv(
    r"C:\Users\architmehta\Telco_Churn_Project\Data\forSQL.csv"
)

print(df.head())
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection
0	7590-VHVEG	Female	0.0	Yes	No	1.0	No	No phone service	DSL	No	...	No
1	5575-GNVEE	Male	0.0	No	No	34.0	Yes	No	DSL	Yes	...	Yes
2	3668-QPYBK	Male	0.0	No	No	2.0	Yes	No	DSL	Yes	...	No
3	7795-CFOCH	Male	0.0	No	No	45.0	No	No phone service	DSL	Yes	...	Yes
4	9237-HQITU	Female	0.0	No	No	2.0	Yes	No	Fiber optic	No	...	No

	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling
0	No	No	No	Month-to-month	Yes
1	No	No	No	One year	No
2	No	No	No	Month-to-month	Yes
3					
4					

Step 2: Explore Dataset Structure

```

print("Shape:")
print(df.shape)

print("\nColumns:")
print(df.columns)

print("\nData Types:")
print(df.dtypes)

print("\nInfo:")
print(df.info())

```

```

Shape:
(7047, 21)

Columns:
Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
       'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
       'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
       'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
      dtype='object')

Data Types:
customerID      object
gender          object
SeniorCitizen   float64
Partner         object
Dependents      object
tenure          float64
PhoneService    object
MultipleLines   object
InternetService object
OnlineSecurity  object
OnlineBackup    object
DeviceProtection object
TechSupport     object

```

Step 3: Check Missing Values

```
print("Missing Values:")
print(df.isnull().sum())
```

Missing Values:

customerID	4
gender	3
SeniorCitizen	4
Partner	4
Dependents	4
tenure	4
PhoneService	4
MultipleLines	4
InternetService	4
OnlineSecurity	4
OnlineBackup	4
DeviceProtection	4
TechSupport	4
StreamingTV	4
StreamingMovies	4
Contract	4
PaperlessBilling	4
PaymentMethod	4
MonthlyCharges	4
TotalCharges	4
Churn	4

dtype: int64

Step 4: Clean the Data

```
# Remove completely empty rows
```


Step 6: Customer Segmentation

```
df['CustomerSegment'] = np.where(
    df['tenure'] < 12,
    'New Customer',
    'Existing Customer'
)

print(df[['tenure', 'CustomerSegment']].head())

print(df['CustomerSegment'].value_counts())
```

Python

	tenure	CustomerSegment
0	1.0	New Customer
1	34.0	Existing Customer
2	2.0	New Customer
3	45.0	Existing Customer
4	2.0	New Customer

```
CustomerSegment
Existing Customer    4974
New Customer        2058
Name: count, dtype: int64
```

Step 7: Analyze Churn by Customer Segment

```
segment_churn = pd.crosstab(
    df['CustomerSegment'],
    df['Churn'],
    normalize='index'
) * 100

print(segment_churn)
```

Python

Step 11: Export Clean Dataset

```
df.to_csv(  
    r"C:\Users\architmehta\Telco_Churn_Project\outputs\cleaned_telco.csv",  
    index=False  
)  
  
print("Cleaned dataset saved successfully!")
```

Python

Cleaned dataset saved successfully!

Step 12: Business Insights

```
print("Business Insights\n")  
  
print(  
    "Highest Churn Customer Segment:",  
    segment_churn['Yes'].idxmax()  
)  
  
print(  
    "Highest Churn Contract Type:",  
    contract_churn['Yes'].idxmax()  
)  
  
print(  
    "Highest Churn Payment Method:",  
    payment_churn['Yes'].idxmax()  
)  
  
print(  
    "Highest Churn Internet Service:",  
    internet_churn['Yes'].idxmax()  
)
```

Step 13: Show Churn Percentages in Insights

```
print(
    f"Customer Segment: {segment_churn['Yes'].idxmax()} "
    f"({segment_churn['Yes'].max():.2f}%"
)

print(
    f"Contract Type: {contract_churn['Yes'].idxmax()} "
    f"({contract_churn['Yes'].max():.2f}%"
)

print(
    f"Payment Method: {payment_churn['Yes'].idxmax()} "
    f"({payment_churn['Yes'].max():.2f}%"
)

print(
    f"Internet Service: {internet_churn['Yes'].idxmax()} "
    f"({internet_churn['Yes'].max():.2f}%"
)
```

Customer Segment: New Customer (48.54%)
Contract Type: Month-to-month (42.71%)
Payment Method: Electronic check (45.29%)
Internet Service: Fiber optic (41.89%)

Business Conclusion:-

Based on the churn analysis, several key patterns were identified:

1. New Customers have the highest churn rate compared to existing customers. This indicates that customers are more likely to leave during their first year of service. Improving onboarding and early customer engagement could help reduce churn.
2. Month-to-Month Contract customers show the highest churn rate among all contract types. Customers with longer contracts (one-year and two-year) tend to be more loyal and less likely to leave.
3. Customers using Electronic Check as their payment method have the highest churn rate. This may indicate lower customer commitment or a less convenient payment experience compared to automatic payment methods.
4. Fiber Optic Internet customers have the highest churn rate among internet service categories. Although Fiber Optic provides faster internet speeds, customers may be dissatisfied with pricing, service quality, or customer support.

Recommendations:-

1. Focus retention efforts on new customers, especially during the first 12 months.
2. Encourage customers to switch from month-to-month contracts to longer-term contracts through discounts and loyalty programs.
3. Promote automatic payment methods such as bank transfer or credit card auto-pay.
4. Investigate the reasons behind the high churn rate among Fiber Optic customers and improve service quality and customer satisfaction.

Final Conclusion:-

1. Customer churn is primarily driven by:
2. Short customer tenure
3. Month-to-month contracts
4. Electronic check payment method
5. Fiber Optic internet service

By targeting these high-risk customer groups with retention strategies, the company can reduce churn, improve customer satisfaction, and increase long-term revenue.

```
CREATE DATABASE telco_project;
USE telco_project;
CREATE TABLE telco_churn
(
  customerID VARCHAR(50),
  gender VARCHAR(20),
  SeniorCitizen INT,
  Partner VARCHAR(10),
  Dependents VARCHAR(10),
  tenure INT,
  PhoneService VARCHAR(10),
  MultipleLines VARCHAR(30),
  InternetService VARCHAR(30),
  OnlineSecurity VARCHAR(30),
  OnlineBackup VARCHAR(30),
  DeviceProtection VARCHAR(30),
  TechSupport VARCHAR(30),
  StreamingTV VARCHAR(30),
  StreamingMovies VARCHAR(30),
  Contract VARCHAR(30),
  PaperlessBilling VARCHAR(10).
```

```
FROM telco_churn
GROUP BY Churn;

SELECT
    Churn,
    ROUND(AVG(tenure),2) AS avg_tenure
FROM telco_churn
GROUP BY Churn;

SELECT
    PaymentMethod,
    COUNT(*) AS total_customers,
    SUM(CASE WHEN Churn='Yes' THEN 1 ELSE 0 END) AS churned_customers
FROM telco_churn
GROUP BY PaymentMethod
ORDER BY churned_customers DESC;

SELECT
    InternetService,
    COUNT(*) AS total_customers,
    SUM(CASE WHEN Churn='Yes' THEN 1 ELSE 0 END) AS churned_customers
FROM telco_churn
GROUP BY InternetService;
```



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7.046K

Total Customers

1.869K

Churned Customers

26.53%

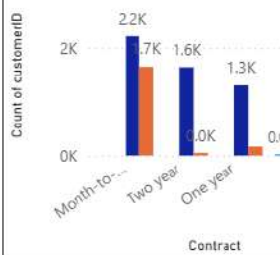
Churn Rate %

64.76

Avg Monthly Charges

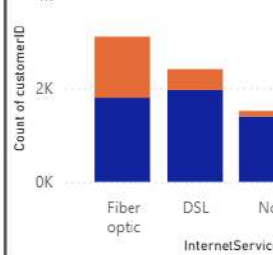
customer churn by Contract

Churn ● No ● Yes



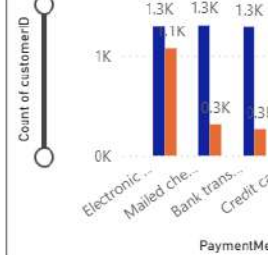
Churn By Internet Services

Churn ● No ● Yes

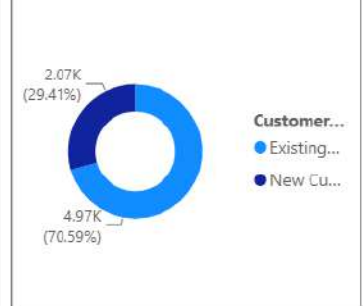


Churn By Payment Method

Churn ● No ● Yes



CustomerSegment



Contract

- ☐ Month-to-month
- ☐ One year
- ☐ Two year

InternetService

- ☐ DSL
- ☐ Fiber optic
- ☐ No

gender

- ☐ Female
- ☐ Male

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
customerID	gender	SeniorCitiz	Partner	Dependent	tenure	PhoneServ	MultipleLir	InternetSe	OnlineSect	OnlineBac	DevicePro	TechSuppc	Streaming	Streaming	Contract	PaperlessE	PaymentM	MonthlyCh	TotalCharg	Churn
7590-VHV	Female	0	Yes	No	1	No	No phone	DSL	No	Yes	No	No	No	No	Month-to-	Yes	Electronic	29.85	29.85	No
5575-GNV	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	No	No	No	One year	No	Mailed che	56.95	1889.5	No
3668-QPY	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to-	Yes	Mailed che	53.85	108.15	Yes
7795-CFO	Male	0	No	No	45	No	No phone	DSL	Yes	No	Yes	Yes	No	No	One year	No	Bank trans	42.3	1840.75	No
9237-HQI	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No	No	No	No	Month-to-	Yes	Electronic	70.7	151.65	Yes
9305-CDS	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	Month-to-	Yes	Electronic	99.65	820.5	Yes
1452-KIO	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No	No	Yes	No	Month-to-	Yes	Credit card	89.1	1949.4	No
6713-OKO	Female	0	No	No	10	No	No phone	DSL	Yes	No	No	No	No	No	Month-to-	No	Mailed che	29.75	301.9	No
7892-POO	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	No	Yes	Yes	Yes	Yes	Month-to-	Yes	Electronic	104.8	3046.05	Yes
6388-TAB	Male	0	No	Yes	62	Yes	No	DSL	Yes	Yes	No	No	No	No	One year	No	Bank trans	56.15	3487.95	No
9763-GRS	Male	0	Yes	Yes	13	Yes	No	DSL	Yes	No	No	No	No	No	Month-to-	Yes	Mailed che	49.95	587.45	No
7469-LKB	Male	0	No	No	16	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	Two year	No	Credit card	18.95	326.8	No
8091-TTV	Male	0	Yes	No	58	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	One year	No	Credit card	100.35	5681.1	No
0280-XIG	Male	0	No	No	49	Yes	Yes	Fiber optic	No	Yes	Yes	No	Yes	Yes	Month-to-	Yes	Bank trans	103.7	5036.3	Yes
5129-JLP	Male	0	No	No	25	Yes	No	Fiber optic	Yes	No	Yes	Yes	Yes	Yes	Month-to-	Yes	Electronic	105.5	2686.05	No
3655-SNQ	Female	0	Yes	Yes	69	Yes	Yes	Fiber optic	Yes	Yes	Yes	Yes	Yes	Yes	Two year	No	Credit card	113.25	7895.15	No
8191-XWS	Female	0	No	No	52	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	One year	No	Mailed che	20.65	1022.95	No
9959-WOF	Male	0	No	Yes	71	Yes	Yes	Fiber optic	Yes	No	Yes	No	Yes	Yes	Two year	No	Bank trans	106.7	7382.25	No
4190-MFL	Female	0	Yes	Yes	10	Yes	No	DSL	No	No	Yes	Yes	No	No	Month-to-	No	Credit card	55.2	528.35	Yes
4183-MYF	Female	0	No	No	21	Yes	No	Fiber optic	No	Yes	Yes	No	No	Yes	Month-to-	Yes	Electronic	90.05	1862.9	No
8779-QRD	Male	1	No	No	1	No	No phone	DSL	No	No	Yes	No	No	Yes	Month-to-	Yes	Electronic	39.65	39.65	Yes
1680-VDC	Male	0	Yes	No	12	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	One year	No	Bank trans	19.8	202.25	No
1066-JKSG	Male	0	No	No	1	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	Month-to-	No	Mailed che	20.15	20.15	Yes
3638-WEA	Female	0	Yes	No	58	Yes	Yes	DSL	No	Yes	No	Yes	No	No	Two year	Yes	Credit card	59.9	3505.1	No
6322-HRP	Male	0	Yes	Yes	49	Yes	No	DSL	Yes	Yes	No	Yes	No	No	Month-to-	No	Credit card	59.6	2970.3	No
6865-JZNK	Female	0	No	No	30	Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to-	Yes	Bank trans	55.3	1530.6	No
6467-CHF	Male	0	Yes	Yes	47	Yes	Yes	Fiber optic	No	Yes	No	No	Yes	Yes	Month-to-	Yes	Electronic	99.35	4749.15	Yes
8665-UTD	Male	0	Yes	Yes	1	No	No phone	DSL	No	Yes	No	No	No	No	Month-to-	No	Electronic	30.2	30.2	Yes
5248-YGIJ	Male	0	Yes	No	72	Yes	Yes	DSL	Yes	Yes	Yes	Yes	Yes	Yes	Two year	Yes	Credit card	90.25	6369.45	No
8773-HHU	Female	0	No	Yes	17	Yes	No	DSL	No	No	No	No	Yes	Yes	Month-to-	Yes	Mailed che	64.7	1093.1	Yes
3841-NFEC	Female	1	Yes	No	71	Yes	Yes	Fiber optic	Yes	Yes	Yes	Yes	No	No	Two year	Yes	Credit card	96.35	6766.95	No
4090-XIHW	Male	1	Yes	No	2	Yes	No	Fiber optic	No	No	Yes	No	Yes	Yes	Month-to-	Yes	Credit card	95.5	181.65	No